Microsoft

# BUILD

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# Open-Source Models

MICROSOFT BUILD 2026 · EXPERT MEET-UP · STATION 3

Microsoft

# Sessions

| | | | |
|---|---|---|---|
| BRK232 | [Train and deploy custom OSS reasoning models with Foundry]() | Wed, Jun 3<br>2:45 PM - 3:30 PM PDT | Open-source reasoning models work out of the box, but production requires domain-specific training. In this session, ML engineers will learn how to use Microsoft Foundry to train and tune open-source reasoning models in a code-first workflow using curated RL environments and modern frameworks. |
| DEM321 | [Train and deploy custom reasoning models using reinforcement learning]() | Tue, Jun 2<br>2:10 PM - 2:35 PM PDT | Go beyond prompt engineering to build custom reasoning models using reinforcement learning in Microsoft Foundry. We'll walk through how to train and fine-tune a model to improve reasoning quality, deploy it into Foundry, and integrate it into an agent workflow. |
| DEM320 | [Hugging Face open-source models to production on Microsoft Foundry]() | Wed, Jun 3<br>10:30 AM - 10:55 AM PDT | Open-source models fuel modern AI but running them in production is hard. In this lightning talk, Microsoft and Hugging Face demonstrate how to deploy and scale Hugging Face models using Foundry Managed Compute inside Microsoft Foundry. |
| LGT419 | [Turn ideas into AI applications with Microsoft Foundry Labs]() | Tue, Jun 2<br>12:20 PM - 12:35 PM PDT | Developers need a fast path from ideas to real AI applications. In this lightning talk, Microsoft introduces Foundry Labs 2.0, a new experience for rapid AI prototyping. Through a live demo, developers will explore domain specific models and data, experiment in interactive playgrounds, and compose models, data, and agents into a working application. |

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# Key Announcements

1. ***Managed GPU compute (no VM or cluster management):*** Preview
2. ***Custom/Code First Model Training with Foundry:*** Demo
3. ***Bring Your Own Weights:*** Demo
4. ***Hugging Face models on Microsoft Foundry***
5. ***Microsoft OSS models on Foundry***

# What is Foundry Managed Compute

Introducing

Microsoft Foundry · April 2026

Private preview

# Managed Compute

## A managed GPU compute platform within Microsoft Foundry

Foundry Managed Compute lets developers deploy, customize, and scale open-source models on dedicated GPU compute, without managing VMs, clusters, or infrastructure. It extends Foundry's model hosting with a new deployment type built specifically for OSS models.

## Where it fits in Foundry

| Pay-per-token | PTU | New<br>Managed Compute |
| --- | --- | --- |
| Serverless token-based inference for 1P models | Reserved throughput units for consistent performance | Dedicated GPU compute for OSS models |

# Built on three core capabilities

## 1 Model Catalog

- Thousands of OSS models from Hugging Face, NVIDIA, and Microsoft Research
- New models published within hours of release
- Enterprise-ready: Vulnerability scanning, licensing compliance, and SafeTensor formats
- Domain and industry-specific providers included

## 2 Optimized Inference Stack

- High-performance runtimes: vLLM, SGLang, and NVIDIA NIMs
- Full weights and LoRA adapter support
- Advanced serving: speculative decoding, disaggregated prefill/decode, prefix caching, tensor parallelism
- Continuous batching for maximum throughput

## 3 Managed GPUs

- Deploy by model instance — no VM or cluster management
- Scale to zero when idle; auto or manual scaling
- Microsoft manages runtimes, scaling, updates, and security patching
- Global and Data Zone deployments for residency needs

# Start with OSS model discovery and baseline evals

Managed Compute turns catalog models into trainable, evaluatable Foundry endpoints.

OSS Catalog → 1-click deploy on Managed Compute → Baseline evals → Traces & failures → SFT / RL / post-training → Redeploy

### Catalog breadth

Thousands of OSS models in Foundry.

Hugging Face, NVIDIA, Microsoft Research, ecosystem partners.

Filter by task, license, publisher.

### One-click deployment

Onto Foundry-managed GPU accelerators…

…or onto your own Azure VM capacity.

Same Foundry experience, either way.

### Same endpoint & auth

Existing Foundry keys and SDK work unchanged.

Private endpoints and portal stay the same.

Standard inference URL per deployment.

### Feeds the loop

Baseline for evals and traces.

Source of synthetic data & preference signals.

Inputs for SFT, RL, and reward workflows.

# Managed Compute — the OSS execution substrate in Foundry

One stack for open models: discover, run, and serve through the same Foundry experience.

| | | |
|---|---|---|
| **Catalog**<br>What you start with | **Model breadth**<br>Thousands of OSS models — Hugging Face, NVIDIA, Microsoft Research, partners. | **Optimized runtimes**<br>vLLM, SGLang, NVIDIA NIM — full weights and LoRA adapters. |
| **Compute**<br>How it runs | **Foundry-managed accelerators**<br>One-click deploy — Foundry owns quota and capacity. | **Bring your own Azure VM capacity**<br>Reuse your training cluster — your fleet, your quota. |

| | | | |
|---|---|---|---|
| **Inference**<br>How you consume | **Same endpoint · same auth · same SDK · wired into evals, agents, observability** | | |
| | **Pay-as-you-go** | **PTU** | New<br>**Managed Compute** |

# Billing and deployment model

## How Customers Pay

- Hourly metered per accelerator SKU (A100, H100, H200, MI300)
- Throughput per GPU as the billing unit — aligned with parameter count and industry benchmarks, enabling direct comparison with alternatives
- Commitment discount TBD

## How Deployments are Sized

- System automatically picks GPUs per model instance based on model size and architecture — no manual VM math
- Customers configure flexible operating points: throughput, latency, or context length per deployment
- Auto-scale and scale-to-zero keep costs aligned with actual traffic

## Quota & Regions

- Quota requested through the same Foundry quota process as pay-per-token and PTU — not Azure Compute quota
- Quota is per SKU per region
- Launching Global initially · Data Zone and regional TBD

# Custom models on Managed Compute

What you bring, what it becomes, where it runs.

| | | | |
|---|---|---|---|
| **1** **Custom models**<br>Where they come from | **Upload from your environment**<br>Drag-and-drop weights from your laptop or storage. | **Register from a training job**<br>Promote an FT job output straight to the registry. | **Import from Hugging Face**<br>Pull any compatible repo into Foundry. |
| **2** **Formats**<br>What you bring | **Full weights**<br>Complete model checkpoint. | **LoRA adapters**<br>Lightweight adapter on a registered base — hot-swappable. | |
| **3** **Assets**<br>What you ship | **BYOW — Bring Your Own Weights**<br>Just the weights; Foundry picks the runtime. | **BYOC — Bring Your Own Container**<br>Your serving image; weights mounted in. | |
| **4** **Compute**<br>Where it runs | **Managed Compute**<br>Foundry-managed GPU or your training cluster. | **Fireworks (PTU)**<br>Provisioned partner inference. | |

# Why build with Hugging Face on Microsoft Foundry

# The latest open models, in a secure environment

New frontier capabilities — agentic coding, video segmentation, speech, embeddings — land on Hugging Face every week.

Microsoft Foundry brings the Hugging Face open ecosystem into a secure, production-ready experience on Azure — security-screened weights, optimized runtimes, managed endpoints.

**Now with Managed Compute,** it's easier than ever to deploy open models in Foundry!

# The Hugging Face Collection on Microsoft Foundry

## Updated Daily

**11,000 models**

Trending models added daily

Safetensors only, no remote code, security scanned

## Every Modality

**Text, vision, audio and more**

LLMs and VLMs for chat and agents

ASR and speech translation

Embeddings, segmentation, image generation

## Optimized Runtimes

**Best engine for the model**

vLLM and SGLang for LLMs

TEI for embeddings

llama.cpp for CPU

# Now available with Foundry Managed Compute

## Simplified vGPUs

**Just select the Accelerator**

NVIDIA A100 80GB

NVIDIA H100 80GB

AMD MI300X

## Quotas Made Easy

**Foundry specific quotas**

Easy GPU availability

Easy request process

## Simple Infrastructure

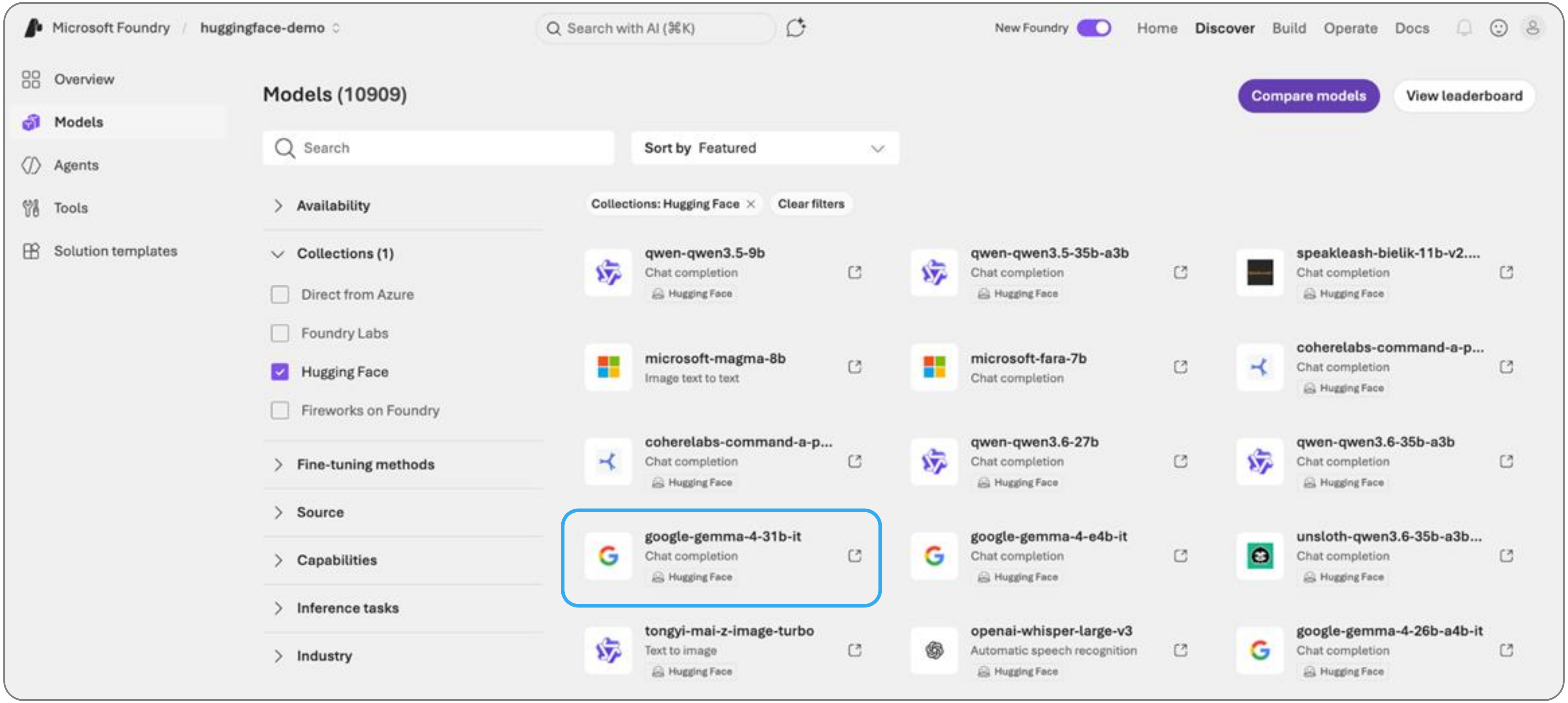
**Just set model replicas**

Auto provisioning

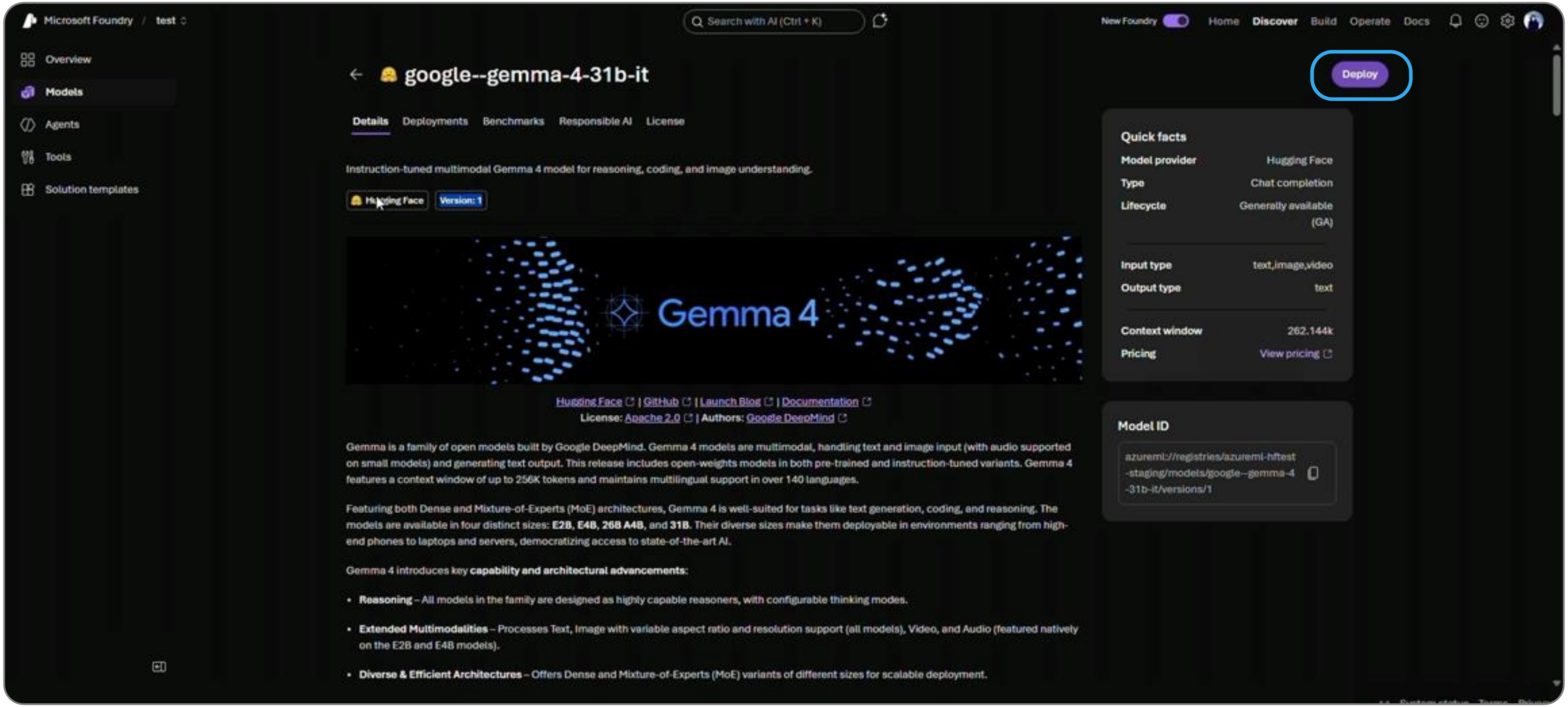
Easy scaling

# How to build with Hugging Face on Microsoft  Foundry

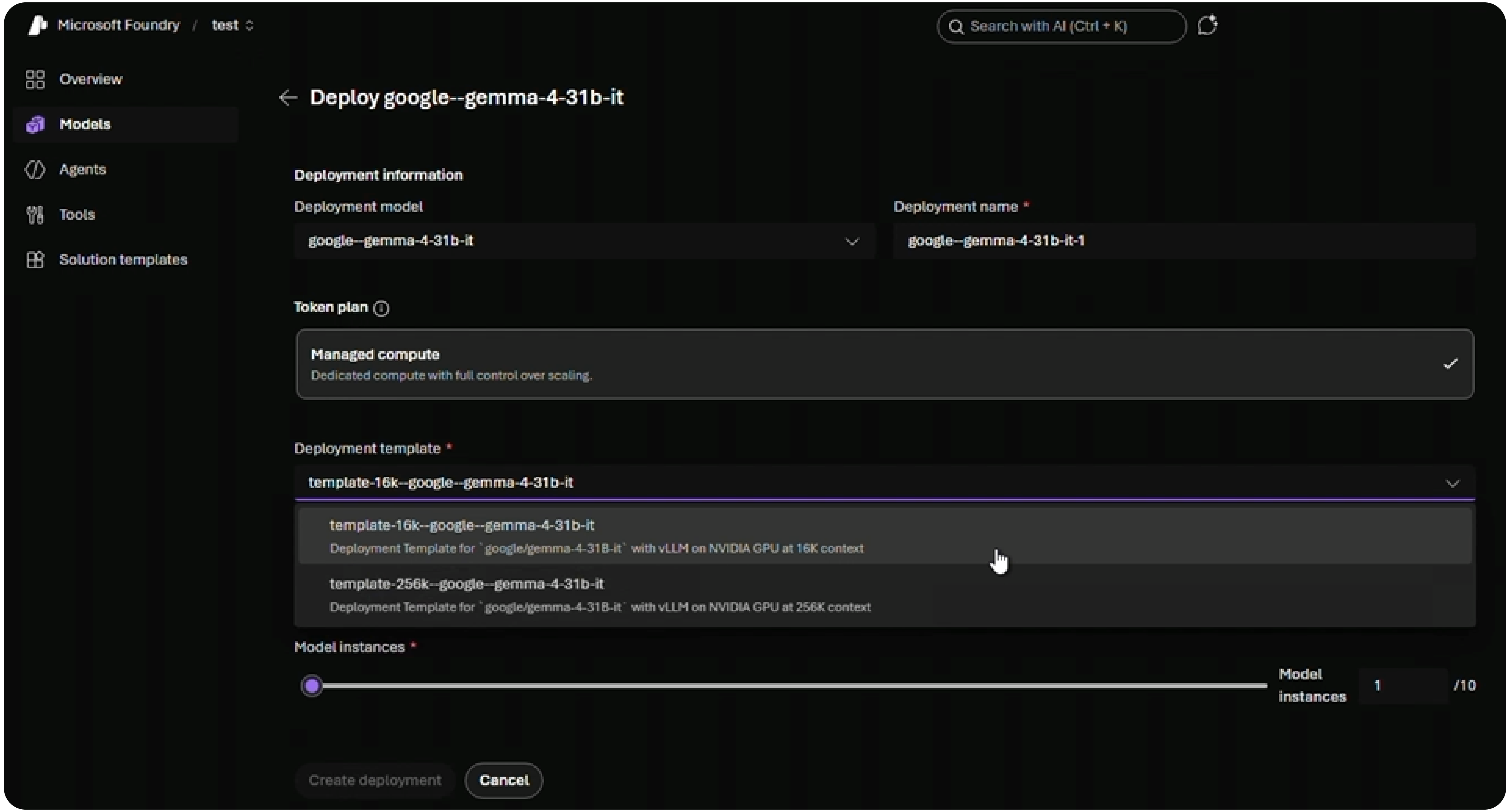
# 1 Select a model in the Hugging Face Collection



# 2 Click deploy

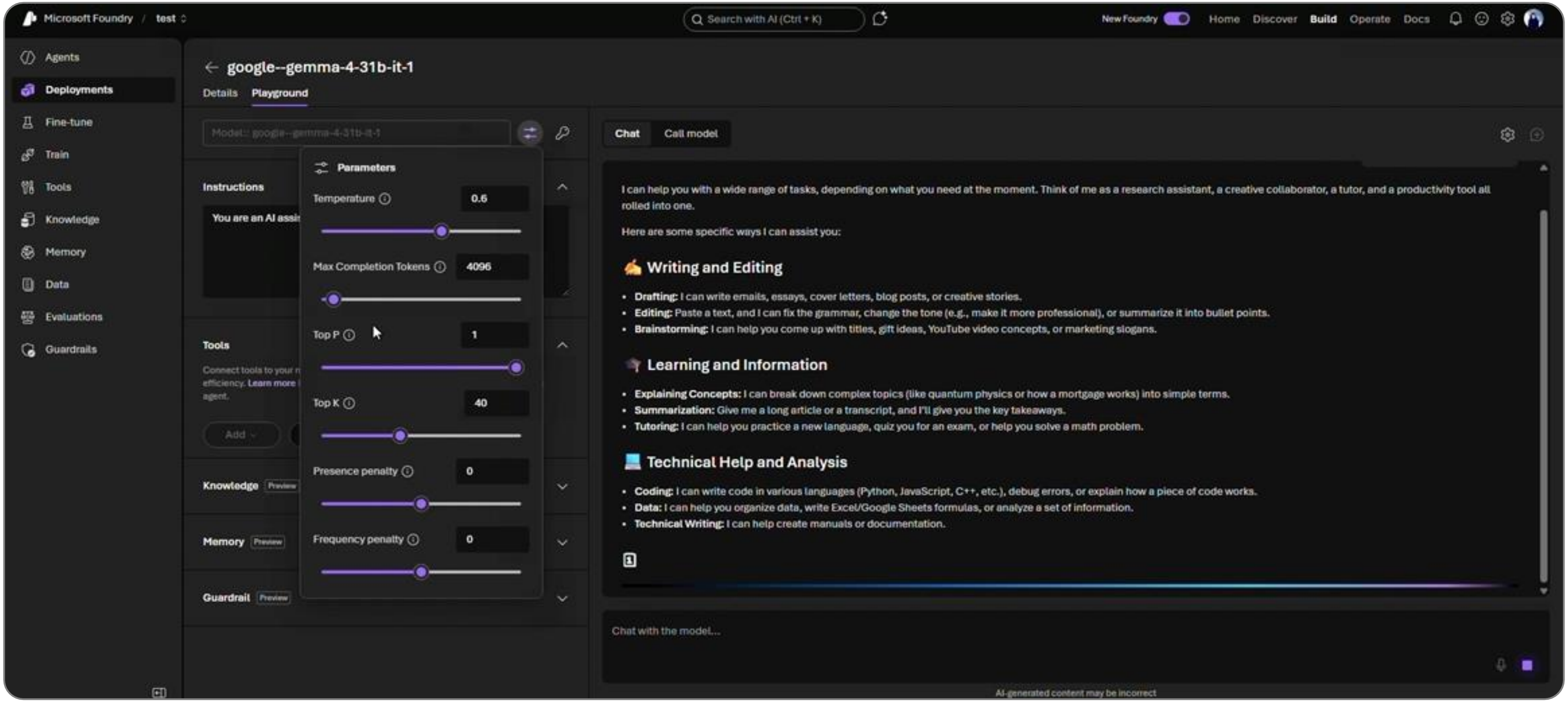


# 3 Select deployment template, accelerator, replicas

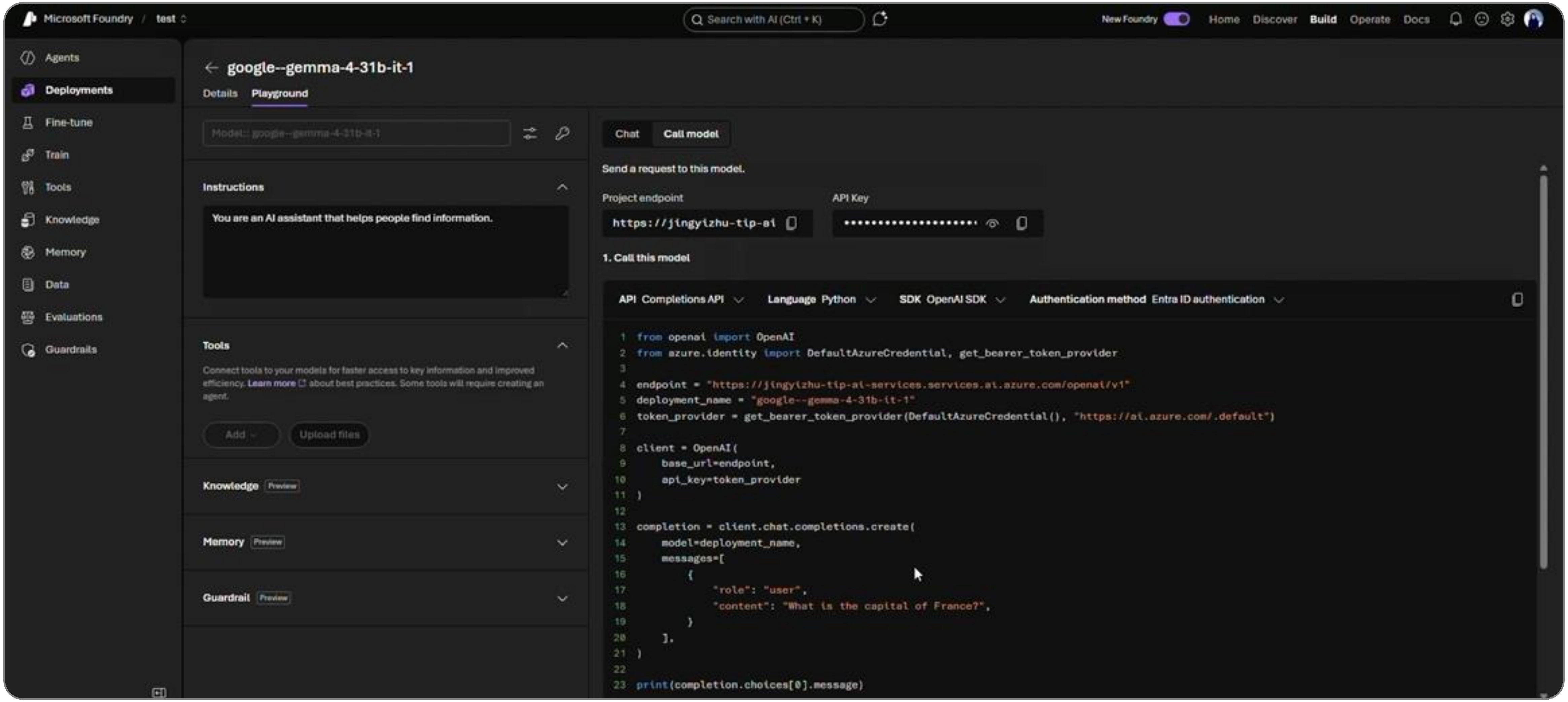


4

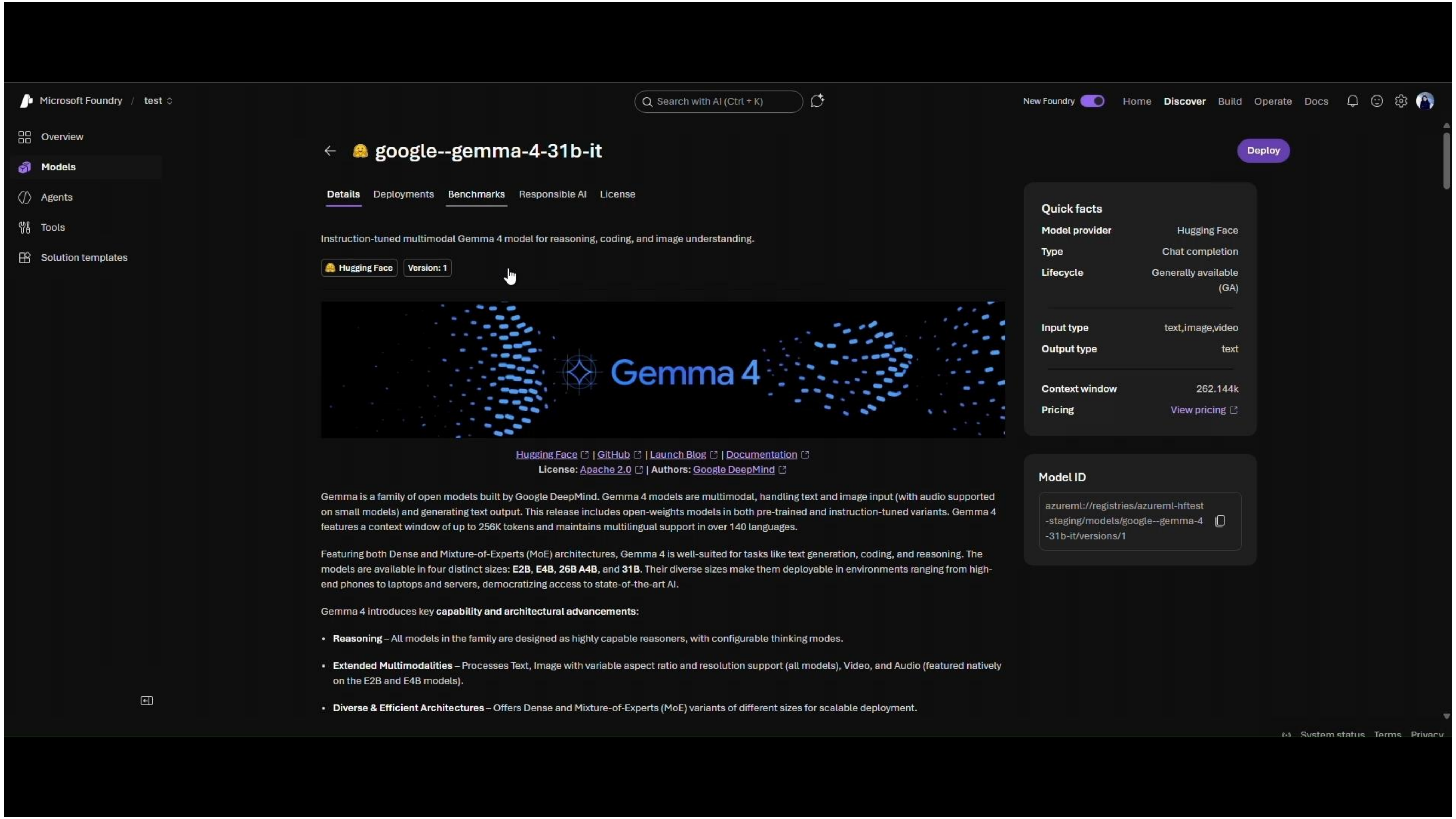
# Test model in Playground



# 5 Integrate managed endpoint

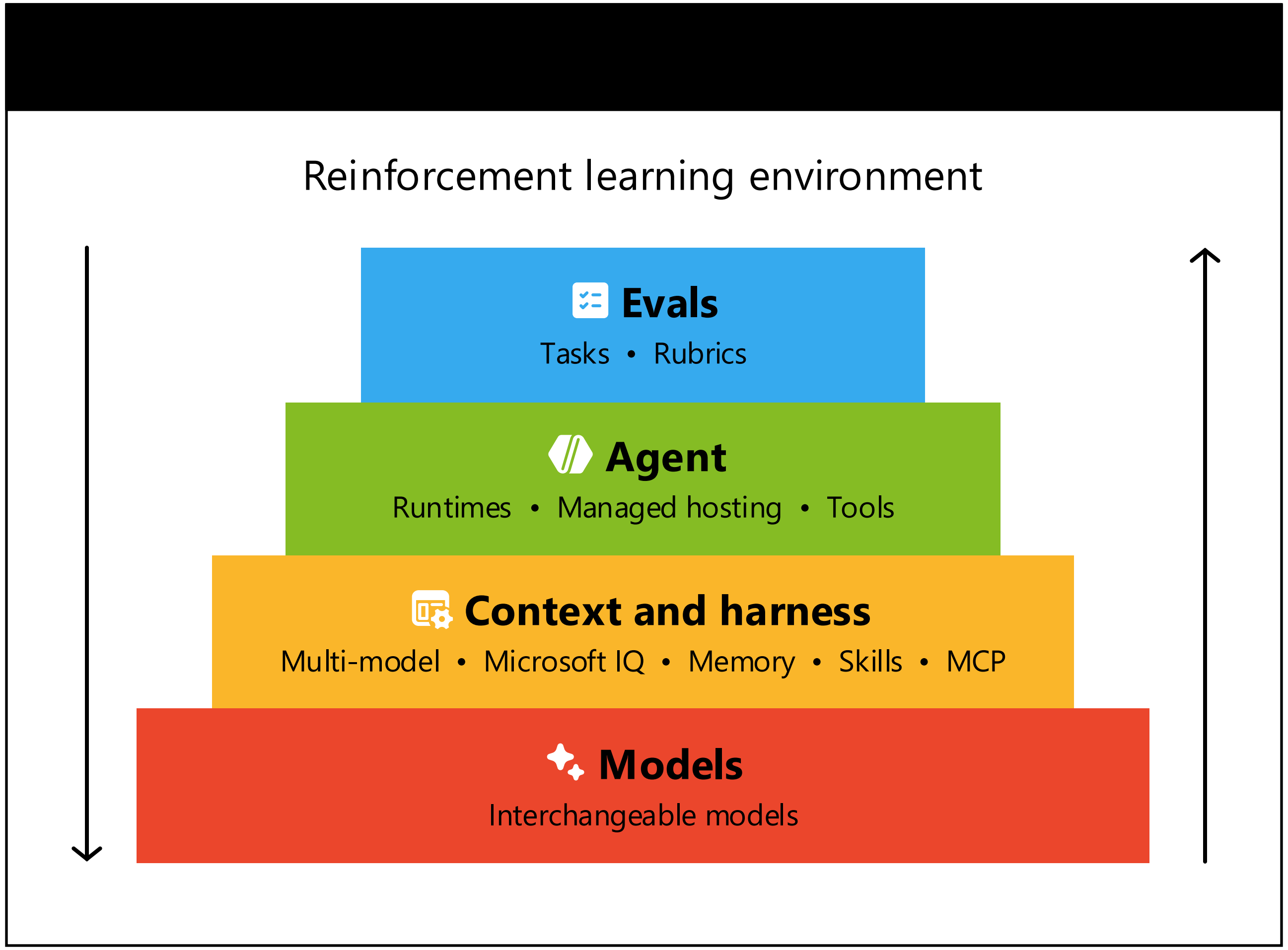


# Managed Compute demo - UI



Microsoft Foundry

# Your hill climbing machine




## Considerations

- Your data, your logs, your weights: all private

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- Developer building blocks: you pick the evals, the harness, and the model – we do the plumbing

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- Optimize for what you need: cost, quality, or latency

LAUNCHING· OSS MODELS ON FOUNDRY

# Improve agent reasoning end to end.

One workflow for training and serving open-source models in Foundry - from first experiment to a smarter agent in production.

01 02 03 04 05

**01**
### Experiment
Pick the model.
Frame the task.

**02**
### Train
SFT, then RFT.
Graders + rewards.

**03**
### Deploy
Managed endpoints.
LLM-native monitoring.

**04**
### Consume in agents
Drop into Foundry agents and tools.

**05**
### Improve
Evaluate. Iterate.
Close the loop.

↺ *every cycle makes the agent smarter*

# Three tiers of training in Foundry

From one click to full contol — same compute, same models, same evals.

**Post-training surfaces · Choose your adventure**

**Foundry-managed** ← → **Self-managed**

**Most teams start here**

1

33%

LOW MEDIUM HIGH

## Managed fine-tuning

Bring your data, get a better model

**WHEN TO USE**

You want the highest likelihood of success with minimal effort. Select a model, pick a method (SFT, DPO, RFT), upload data. Foundry handles the rest.

**FOUNDRY RUNS**

Everything: data prep, training orchestration, monitoring, deployment. Just bring your data.

2

66%

LOW MEDIUM HIGH

## Serverless training API

Serverless RL with programmatic control

**WHEN TO USE**

You need control over rewards & hyperparameters but don’t’ want to manage infrastructure. Produces LoRA adapters.

**FOUNDRY RUNS**

Serverless backend: rollout, scoring, distributed execution, checkpointing. No cluster management.

3

100%

LOW MEDIUM HIGH

## Full control RL

Distributed RL training on your terms

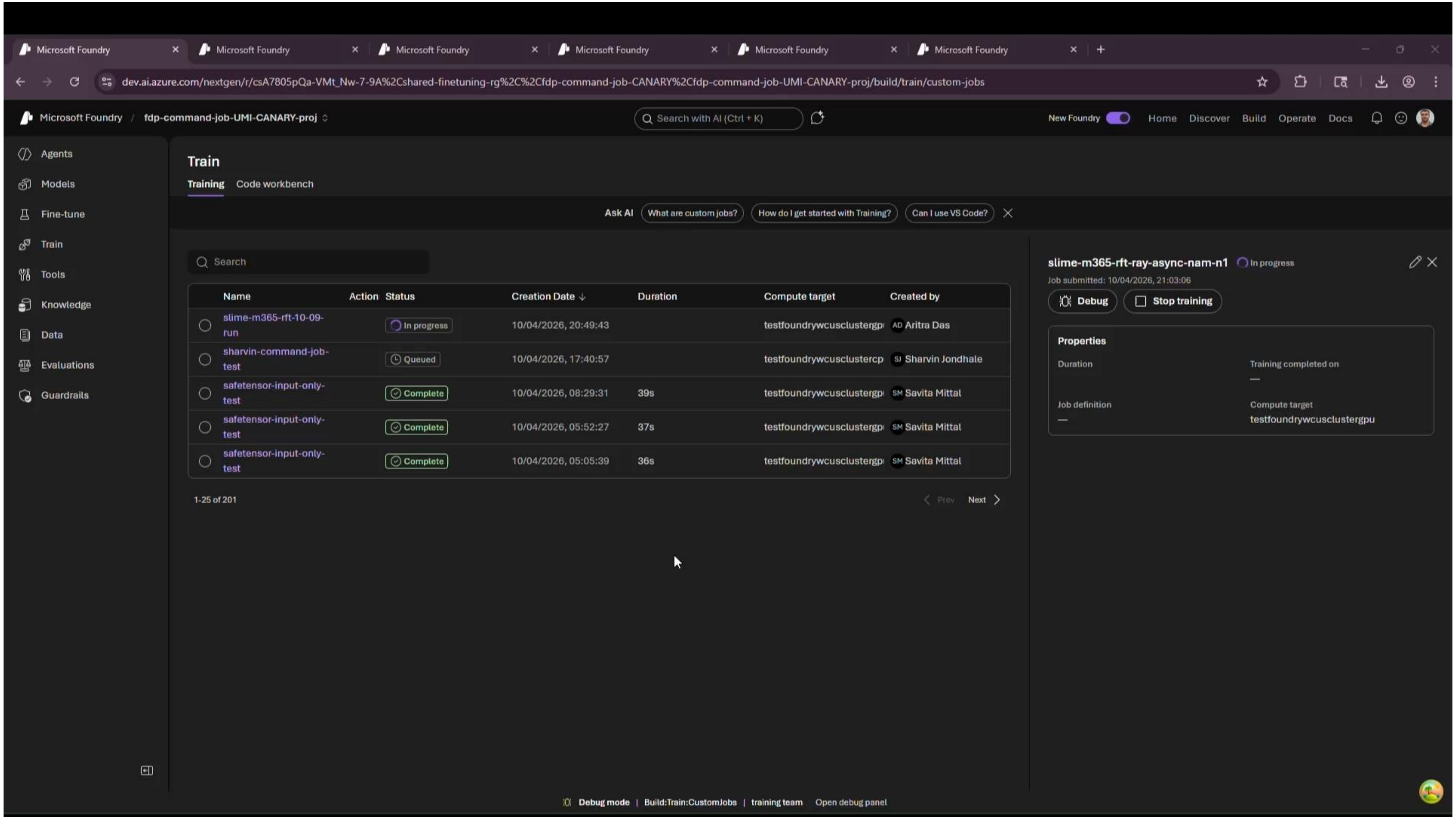
**WHEN TO USE**

You need the full training stack: novel algorithms, custom distributed strategies, large GPU clusters. Your choice of TRL, Slime, or VERL.

**FOUNDRY RUNS**

Managed compute, cluster orchestration, monitoring. You own the code.

# Custom Training Job -  UI



# Bring your own weights — Custom Models

From model selection to live endpoint, no VMs, no cluster setup, no infrastructure expertise required.

01 **Import your model**
Upload from local storage or register from a training job

02 **Find a matching base model from the Catalog**
Foundry maps your model's base lineage to compatible GPU + serving framework configurations automatically

03 **Select a GPU family**
Pick A100, H100, H200, or MI300 — no VM SKU knowledge needed

04 **Configure scaling**
Set auto-scale or manual scale; choose scale-to-zero idle timeout

05 **Deploy and infer**
Your model is live with a standard Foundry endpoint, ready for inference

**Available GPU families**

- **A100 (80GB)**
  Baseline compute — production-grade inference
- **H100**
  Scale-up compute — high-throughput workloads
- **H200**
  Next-generation performance
- **MI300**
  AMD alternative for generative inference

# Train custom models anywhere, deploy and scale in Foundry

Train in Foundry or bring your existing model weights | Leverage cutting-edge speculative decoding and optimized inference with fully custom containers

**01 WEIGHTS** shared

Upload from your environment · Register from a training job · Import from Hugging Face

## BYOW — Bring Your Own Weights

**02** RUNTIME

**Choose a base model from the catalog**

Foundry maps lineage to a compatible serving runtime — vLLM, SGLang, NIM.

**03** DEPLOY

**Foundry-managed compute or Fireworks (PTU)**

Foundry-owned capacity, your own Azure VM cluster, or PTU on Fireworks — pick per workload.

## BYOC — Bring Your Own Container (+ weights)

**02** RUNTIME

**Build and register your own image**

Your serving stack and dependencies, pushed to ACR or built from a Foundry base image.

**03** DEPLOY

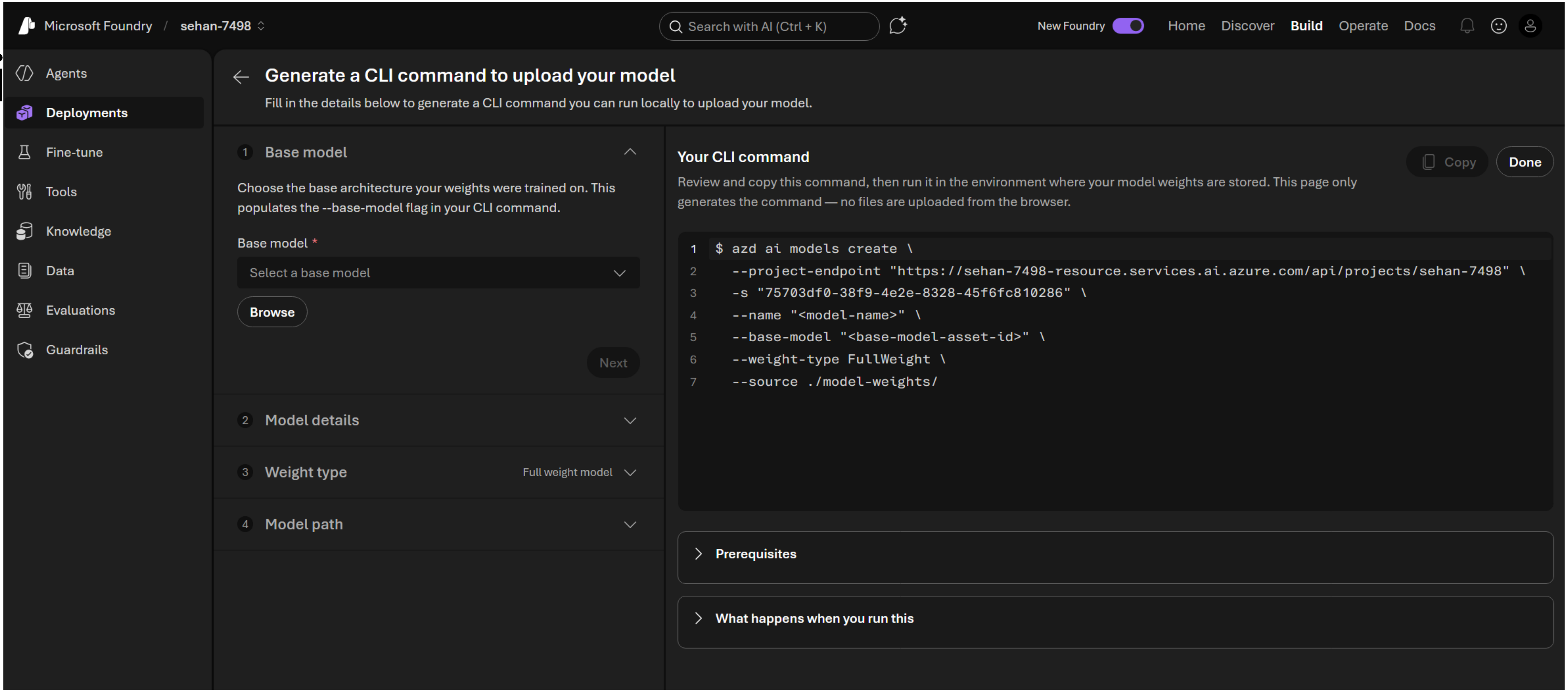
**Your compute cluster only**

Foundry-managed GPU is multi-tenant and serves only catalog runtimes for tenant isolation — BYOC requires your own capacity.

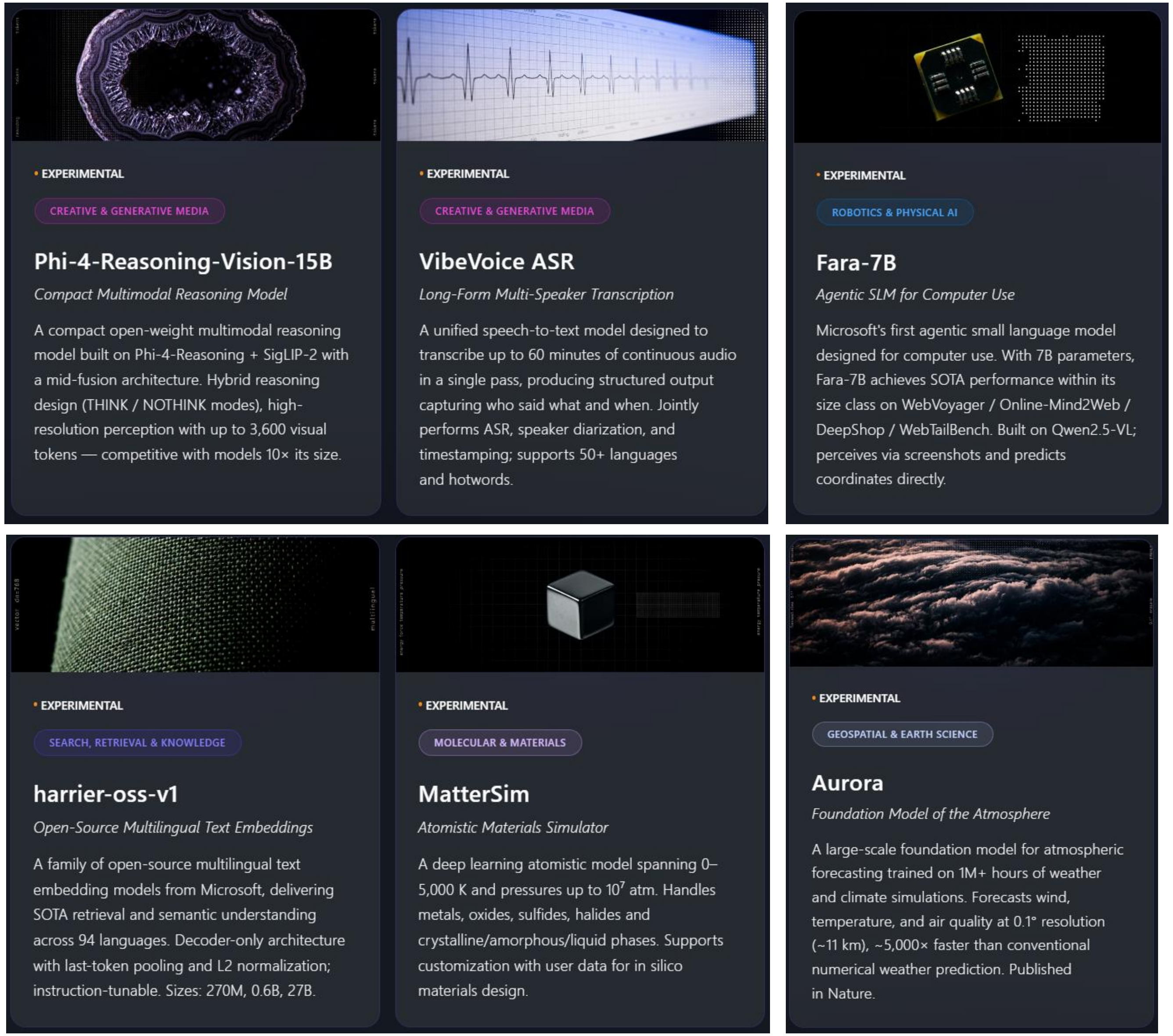
**04 INFERENCE** shared

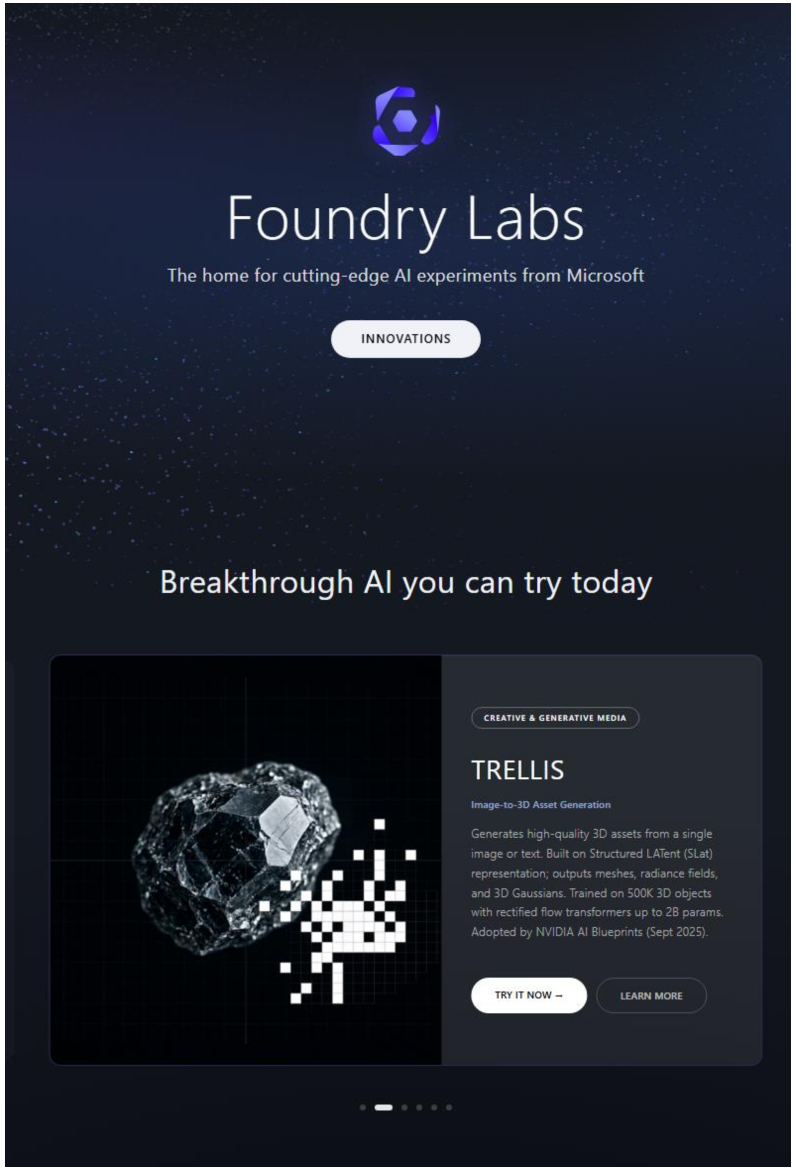
Integrated Foundry endpoint · same auth · same SDK · same evals · same observability

# Bri


Microsoft Foundry / sehan-7498
Search with AI (Ctrl + K)
New Foundry
Home Discover Build Operate Docs
Agents
Deployments
Fine-tune
Tools
Knowledge
Data
Evaluations
Guardrails
Generate a CLI command to upload your model
Fill in the details below to generate a CLI command you can run locally to upload your model.
1 Base model
Choose the base architecture your weights were trained on. This populates the --base-model flag in your CLI command.
Base model *
Select a base model
Browse
Next
2 Model details
3 Weight type
Full weight model
4 Model path
Your CLI command
Copy
Done
Review and copy this command, then run it in the environment where your model weights are stored. This page only generates the command — no files are uploaded from the browser.
$ azd ai models create \
--project-endpoint "https://sehan-7498-resource.services.ai.azure.com/api/projects/sehan-7498" \
-s "75703df0-38f9-4e2e-8328-45f6fc810286" \
--name "<model-name>" \
--base-model "<base-model-asset-id>" \
--weight-type FullWeight \
--source ./model-weights/
Prerequisites
What happens when you run this

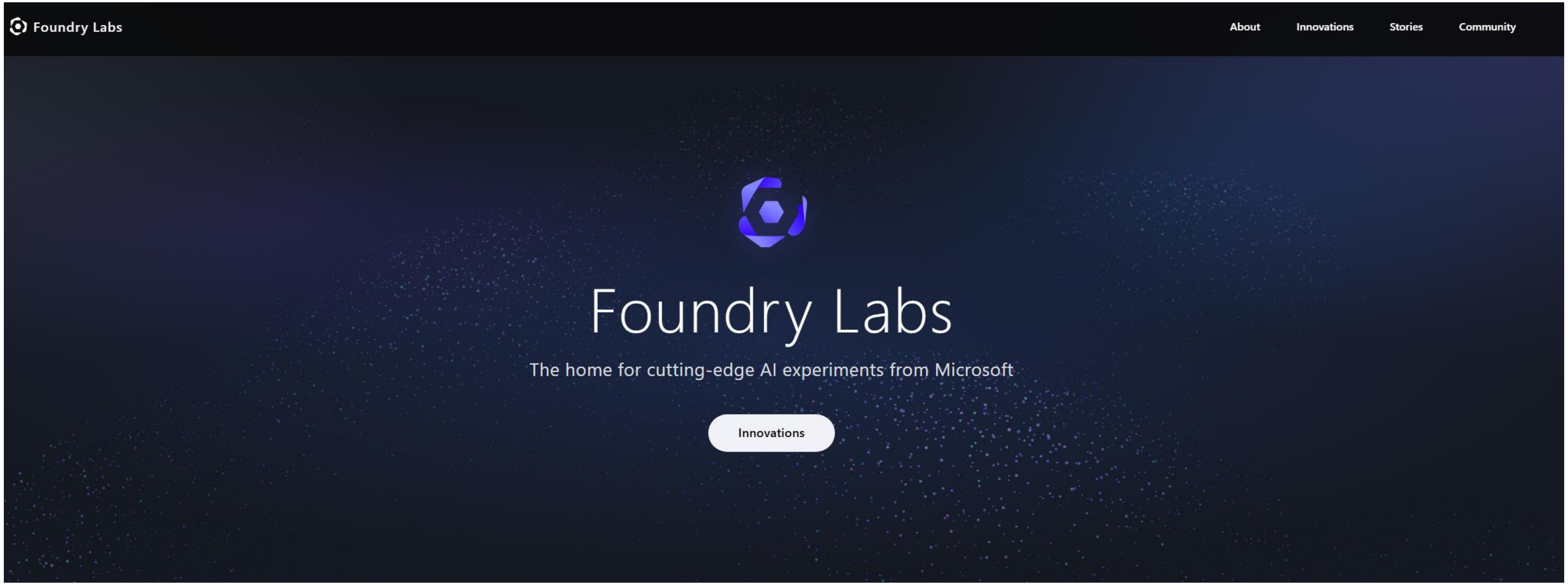
# Foundry Labs



# Demo

Link: [Microsoft Foundry Labs | Early-Stage AI Experiments & Prototypes]



# Thank you!